AMERICAN JOURNAL OF PHYSICAL ANTHROPOLOGY 143:579–590 (2010)

Encephalization, Expensive Tissues, and Energetics: An Examination of the Relative Costs of Brain Size in Strepsirrhines

Nancy L. Barrickman[1]* and Maggie J. Lin[2]

[1]*Department of Anthropology, University of Waterloo, Waterloo, Ontario, Canada N2L 3G1*
[2]*School of Medicine, University of Maryland, Baltimore, MD 21201*

KEY WORDS strepsirrhine; brain size; metabolic rate; intestinal size

ABSTRACT The evolution of encephalization requires that energetic challenges be met. Several hypotheses, such as the maternal energy and expensive tissue hypotheses, have been proposed to explain how some species are able to provide adequate energetic resources for large brains. The former incorporates maternal investment strategies, such as extended life history and elevated resting metabolic rate, which contribute to the growth of a large brain. The latter incorporates the reduction of gut size, which increases available energy for the maintenance of adult brain size. This study examines a sample of strepsirrhines, testing the hypothesis that encephalized species utilize some combination of the above-mentioned strategies. Infants and juveniles from three species at the Duke Lemur Center (DLC) were measured periodically to arrive at head and body growth trajectories. These data were used to determine the energetic tradeoff among the offspring. The examination of gestation length, weaning age, intestinal size and resting metabolic rate was used to assess adult brain maintenance and maternal energetic contribution. The results reveal that *Daubentonia*, the most encephalized and thus human-like of the lemurs, does not experience an energetic trade-off between brain and body during ontogeny, but does exhibit a trade-off between extensive brain growth and possibly reduced intestinal growth. Also, maternal energy is utilized. Encephalized lemurs, such as *Daubentonia*, have higher resting metabolic rate, while encephalized lorisiforms have a longer period of gestation. These results demonstrate that there are several strategies for meeting the energetic demands of encephalization, and they can be manifested differentially across taxa. Am J Phys Anthropol 143:579–590, 2010. © 2010 Wiley-Liss, Inc.

Inquiries into the benefits of encephalization are crucial to understanding the evolution of increased brain size within the primate lineage (Milton, 1981; Byrne, 1988; Sawaguchi, 1988; Sawaguchi, 1990; Barton, 1996; Reader and Laland, 2002; Byrne and Corp, 2004; Deaner et al., 2007; Dunbar, 2003, 2007; Shultz and Dunbar, 2007; Sol et al., 2008; Maclean et al., 2009). However, assessing the costs of evolving a large brain is also imperative to reconstructing the evolution of encephalization. Brains are energetically expensive to grow and maintain, requiring more energy at rest than most other somatic tissues (Kety and Schmidt, 1948; Kety, 1957; Mink et al., 1981; Holliday, 1986). Also, unlike other tissues, the brain cannot withstand temporary energy shortages (Karasov et al., 2004; Bauchinger et al., 2005). Thus, evolving a larger brain size requires that energetic constraints are lifted (Armstrong, 1983; Foley and Lee, 1991; Leonard and Robertson, 1992; Hofman, 1983, 1993; Leonard and Robertson, 1994; Aiello and Wheeler, 1995; Martin, 1983, 1990, 1996, 1998; Leonard et al., 2003; Isler and van Schaik, 2006a,b, 2009; Skoyles, 2008; Martin and Isler, in press). Assessing the evolutionary strategies requires an examination of both growth of the brain and maintenance of the adult brain. Strategies manifested during growth include lengthening of the growth periods and life-history stages, while strategies for adult brain maintenance include increasing resting metabolic rate (RMR) and decreasing the size of other expensive tissues such as the gut. However, higher RMR could also reflect higher maternal contribution to offspring growth, and reduced gut size also indicates less investment in this tissue during growth.

Sacher and Staffeldt (1974) first documented the correlation between the length of gestation and encephalization in mammals. Since then, many studies have found that encephalization is associated with prolonged duration of most life-history stages among mammals and more specifically among primates (Sacher and Staffeldt, 1974; Sacher, 1978; Harvey and Clutton-Brock, 1985; Harvey et al., 1987; Austad and Fischer, 1992; Allman et al., 1993; Allman, 1995; Hakeem et al., 1996; Allman and Hasenstaub, 1999; Barton, 1999; Ross and Jones, 1999; Judge and Carey, 2000; Deaner et al., 2003; Leigh, 2004; Ross, 2003, 2004; Barrickman et al., 2008). In par-

Grant sponsors: Duke University, Graduate Mentorship Grant, the Undergraduate Research Grant, the Molly H. Glander Memorial Undergraduate Research Grant.

*Correspondence to: Dr. Nancy Barrickman, Department of Anthropology, University of Waterloo, PAS Building Room 2012, Waterloo, Ontario, Canada N2L 3G1. E-mail: nlbarric@uwaterloo.ca

Received 25 August 2009; accepted 6 May 2010

DOI 10.1002/ajpa.21354
Published online 8 July 2010 in Wiley Online Library (wileyonlinelibrary.com).

© 2010 WILEY-LISS, INC.

ticular, there is a correlation between greater degrees of encephalization and longer periods gestation among primates, and particularly among anthropoids (Martin, 1990; Barrickman et al., 2008; Martin and Isler, in press). This pattern of extended life history could be enforced by the energetic cost of growing a large brain, as Sacher and Staffeldt (1974) proposed when they argued that the brain was the pacemaker of growth. Specifically, these costs may be reflected in the extended duration of brain growth in species with larger brains, which has cascading effects on all the stages of developmental life history. However, Leigh (2004) analyzed a sample of seven primate species and found no correlation between the duration of brain growth and the age at maturity regardless of whether raw variables or size-adjusted variables were analyzed. In addition, he found a negative correlation between the size-adjusted variables of weaning age and duration of brain growth. However, Leigh (2004) did not test the possibility that brain growth rates could vary as a function of somatic growth rates. Large brain size could be attained by increasing the rate of brain growth relative to the ancestral rate, and as a consequence, species with large brains may exhibit slower somatic growth than that of small-brained primates to compensate for the energetic investment in rapid brain growth (Vinicius, 2005; Barrickman, 2008).

The energetic constraints of brain maintenance have also been the subject of previous research. RMR in adults is crucial factor that could explain how encephalized species are able to afford the costs of a large brain. Early proponents of this idea hypothesized a functional relationship between RMR and brain size, arguing that brain size is constrained by energetic turnover as evidenced by the similar scaling relationships of both variables against body mass (Bauchot, 1978; Eisenberg and Wilson, 1978; Martin, 1981, 1983; Armstrong, 1983, 1985). However, McNab and Eisenberg (1989) found no correlation between the residuals of brain mass on body mass and the residuals of RMR on body mass among mammals. Other studies of the impact of metabolic rate on encephalization have found similar results (Pagel and Harvey, 1988; Barton, 1999), suggesting that species with large brains for their body size do not necessarily have high metabolic rates for their body size. Humans are an ideal example, having a particularly large brain relative to body size but not a remarkably high RMR (Martin, 1998; Martin and Isler, in press). However, Martin (1998) disputed the line-fitting techniques used in McNab and Eisenberg's study. Once the data were reanalyzed, a statistically significant correlation was found. In addition, the rodent taxa, which comprise approximately 40% of the sample, had erroneously high values for brain size. Mace et al. (1981), from which all of the data were obtained, had inadvertently added 0.59 g to the average brain mass of each of the rodent species. Also, the study by McNab and Eisenberg (1989) did not use any method, such as independent contrasts, to control for the effect of phylogeny. Reanalyses, including a sample of 347 mammalian species and corrected data for rodents, demonstrated a significant relationship between RMR relative to body mass and encephalization, using both raw data and independent contrasts (Isler and van Schaik, 2006b). The correlation is also evident among primates (Isler et al., 2008; cf. Barton, 1999).

However, this hypothesis that brain size is governed by RMR remains problematic for several reasons. First, the similarities in scaling relationships and the correlations between size-adjusted variables do not necessarily demonstrate a causal connection in either direction. Second, Martin (1998) noted that the variation in brain size relative to body size greatly exceeds the degree of variation in RMR relative body size (Martin and Isler, in press), which indicates that a great deal of variation in encephalization remains unexplained. Finally, while the analyses by Isler and van Schaik (2006b) and Isler et al. (2008) are statistically significant, the *r*-square values are very low (among mammals: 0.053 and 0.026 in analyses of raw and contrasted variables, respectively; among primates: 0.35 and 0.226 in analyses of raw and contrasted variables, respectively) indicating that only a small amount of the variation in encephalization is accounted for by relative RMR.

The maternal energy hypothesis was proposed to address these difficulties. It merges two of these strategies—metabolic rate and life history—for meeting the energetic demands of a large brain, and considers brain growth in addition to maintenance of the brain during adulthood (Martin, 1996, 1998, 2007; Martin et al., 2005; Martin and Isler, in press). This hypothesis states that the size of the brain is constrained by the amount of energy that a mother can provide during the early stages of her offspring's ontogeny. Increasing the duration of the gestational and lactational period or increasing the energy inputs through higher metabolic turnover to the offspring could provide the necessary energy to grow a large-brained offspring. In analysis of placental mammals, Martin and Isler (in press) found that even though both gestation length and basal metabolic rate show a positive correlation to relative adult brain size when examined separately from each other, they have a negative partial correlation when analyzed together. This result suggests that strategies differ across taxa. Some large-brained species may have longer growth periods whereas others may increase basal metabolic rate, but these factors are not found in combination.

The additional energy needed to maintain a larger brain could also be acquired by reducing the size of other expensive tissues. Along these lines, Aiello and Wheeler (1995) proposed the expensive tissue hypothesis, noting that large brain size relative to body size is significantly correlated with small gastrointestinal tract mass relative to body size among adult anthropoid primates (see also Aiello et al., 2001). Isler and van Schaik (2006a) reanalyzed a larger sample using independent contrasts, and confirmed the significant negative correlation among anthropoids. However, Hladik et al. (1999) argue that gut mass does not accurately reflect the energetic investment. Instead, the surface area of the absorptive mucosa, which receives extensive blood supply and requires a considerable amount of energy per unit area to perform its function, is a better measure of the energetic costs of the gut. When the relationship between surface area relative to body mass and brain mass relative to body mass is examined, humans do not have a remarkably small gut; we have approximately the amount of surface area expected for a frugivorous primate. Yet, gut mass does reflect the thickness of the epithelial tissue and smooth muscle, which also represents a substantial energetic investment, so it is worth considering both mass and surface area. Finally, it is important to note that previous studies of expensive tissue have focused on the maintenance of adult brain tissue. We argue that these analyses are also applicable to brain growth because a reduced gut size indicates that the offspring

will invest less energy in growing its own gut and may be able to shift additional resources to brain growth.

Isler and van Schaik (2009) proposed the expensive brain hypothesis in order to incorporate each of these strategies (Isler and van Schaik, 2006a). They argue that larger-brained species could compensate for these costs in the following ways: (1) the duration of brain growth could be extended, distributing the energetic costs over a longer period of time and thereby decreasing the risk of energy shortages; (2) the RMR could be increased; (3) energy allocation could be shifted by adjusting of the size of other expensive tissues. These trade-offs are not mutually exclusive, and there is likely to be different combinations of the three strategies among encephalized species. In addition, these costs are evident both during the growth of the brain and the maintenance of the brain tissue in adults.

This study examines the various manifestations of these strategies among strepsirrhines. We investigated the costs to the offspring during ontogeny by examining the potential trade-off between brain and body growth in three strepsirrhine species, one of which exhibits a high degree of encephalization (*Daubentonia madagascariensis*). We addressed the maternal costs associated with supporting growing offspring by examining duration of gestation and lactation, the intestinal size, and the RMR in the adults of several strepsirrhines with differing degrees of encephalization. These data were used to determine which, if any, of the above-mentioned strategies are utilized among the members of this taxon. The most encephalized species was expected to have the most prolonged life history and brain growth duration to compensate for the energetic costs of brain growth. In addition, it was expected that large degrees of encephalization would be associated with high RMR and reduced intestinal size to meet the energetic demands of brain maintenance and maternal contributions to offspring brain growth. In addition, reduced intestinal size suggests that the offspring invest less energy in the growth of this tissue compared with brain tissue. Among these species, various combinations of strategies may be employed, and thus all of the variables were analyzed simultaneously. Other studies have addressed similar questions in strepsirrhines (Kappeler, 1996; Rasmussen and Izard, 1988) and found negative results, possibly because they only examined one variable (e.g. life history or RMR) relative to encephalization instead of examining the relative contribution of all of these strategies.

MATERIALS AND METHODS

Growth rates

Data on head growth and somatic growth were collected for three species: *Lemur catta*, *Propithecus coquereli*, and *Daubentonia madagascariensis*. These species were chosen because *D. madagascariensis* is particularly encephalized, *L. catta* is intermediate, and *P. coquereli* has a small brain for its body size (Table 1 and Fig. 1).

Infants from all three species were measured periodically to determine the duration and rate of growth in the cranium, and data on mass were obtained from the Duke Lemur Center (DLC) records. Measurements were taken at one to two month intervals between December 2003 and June 2005, and mass records from September 2003 to September 2005. Adults from each species were also measured to determine the endpoint of growth. In addition, measurements from cadavers were taken to supplement the data gathered from living individuals. The subjects are listed in Table 2. The raw data is too extensive to provide in this article, but is available upon request.

Head size approximates brain size, though the exact relationship in these individual lemurs is not known. Studies have demonstrated a close relationship between brain size and head circumference in humans (Epstein and Epstein, 1978; Cabana et al., 1993). This relationship cannot be translated to lemurs and the measurements used in this study cannot be used to infer absolute growth rates in brain size. However, the studies of humans are suggestive that these measurements can approximate growth rates and durations. Head growth was calculated from measurements of cranial breadth and length. Calipers were placed just above the external auditory meatus to obtain cranial breadth, and above the glabellar region and the most posterior point in the occipital region to obtain cranial length. No measure of cranial height was obtained because none of the animals have a substantial height at any age, nor do they have adequate external landmarks. Cranial circumference was also not taken because there is insufficient cranial height on any of the animals to wrap a measuring tape around the cranium. Head size was calculated as the area of an ellipse, and given that all these lemur species have limited cranial height, this approximation is sufficient to estimate relative rate and durations across the three species included in this study.

Adult brain size and body size

Brain and body mass averages were taken from Isler et al. (2008) or from the records at the Duke Lemur Center (DLC) as noted in Table 1. Some of the values for brain size were measured in units of mass (original sources available in Isler et al., 2008) and others were converted from measures of cranial capacity. Several researchers have examined large datasets of cranial capacity and brain mass in a wide range of taxa to determine whether these two measurements, brain mass and cranial capacity, vary in a systematic fashion. Stephan et al. (1981) found that the specific gravity of primate and insectivore brains is ~1.036. Because cranial capacity exceeds brain volume by approximately the same degree as specific gravity (about 1.04), many researchers have used cranial capacity interchangeably with brain mass (Hofman, 1983; Armstrong, 1985; Falk, 1987). Martin (1990) examined the relationship between brain mass and cranial capacity in a wide range of primates and other mammals, and found a positive allometric relationship. However, Isler et al. (2008) reexamined the relationship between brain mass and cranial capacity in 72 primate species, almost twice as many as Martin (1990) had available. They found that cranial capacity and brain mass are isometric, and the intercept of the regression is very close to the specific gravity of the brain (1.036). Thus, there are no disproportionate changes in brain mass with changes in cranial capacity. All the values of cranial capacity used in this study were converted to brain mass by multiplying the average by the intercept of 1.036.

Values for adult body mass were taken from wild sources whenever possible. The most recent compilations by Isler et al. (2008) and Gordon (2006) include sources from studies in the wild for many of the species included in this study. However, adult body mass from wild ani-

TABLE 1. Averages of brain mass, resting metabolic rate, intestinal mass, and intestinal surface area

Species	Brain mass (g)[a]	Body mass (g)[b]	Gestation length (months)[c]	Weaning age (months)[c]	Resting metabolic rate (VO_2 ml/h)[d]	Body mass for RMR (g)	Intestinal mass (g)[e]	Body mass for intestinal mass (g)	Intestinal surface area (cm^2)[e]	Body mass for intestinal surface area (g)[f]
Daubentonia madagascariensis	46.5	2592	5.5	8.8	1393.9	2562	50.1	2030	490.8	2300
Mirza coquereli	6.0	312	2.0				18.1	328	185.0	293
Microcebus murinus	1.7	65	2.3	1.3	42.3	54	4.6	80	90.7	80
Cheirogaleus medius	2.7	140	4.6	2.0	101.2	259	14.8	125	256.1	199
Cheirogaleus major	6.0	400	4.7	2.3						
Varecia variegata variegata	33.3	3575		2.9	603.6	3512			1282.7	3410
Hapalemur griseus griseus	14.6	709	4.6	4.3			24.0	675	478.2	675
Lemur catta	23.7	2210	4.4	4.7	651.4	2527	70.2	2220	1110.9	2230
Eulemur macao macao	25.4	2390	4.2	4.4					785	1980
Eulemur fulvus	26.7	2292	4.1	5.2			68.9	2053	874.3	2040
Eulemur coronatus	21.4	1180	2.9						899.9	1440
Eulemur mongoz	20.2	1212	3.9	5.0	940	1403			932.6	1310
Propithecus coquereli	31.2	4052	4.6	6.0	1012.3	4052	192.8	3500	2861.1	3500
Propithecus verreauxi	27.1	2955	5.5	6.0	670.0	3350				
Lepilemur leucopus	7.1	606	2.0						456.0	630
Galago demidovii	2.7	75	3.6		59.4	63.8				
Galago moholi	3.8	148	4.0	3.2	198	275				
Galago senegalensis	4.1	194	4.7	3.2	137.2	171.5	12.8	148	105.3	113
Euoticus elegantulus	5.7	274			215.7	261.5				
Otolemur crassicaudatus	12.2	1150	4.4	4.4	523.0	1039				
Otolemur garnetti	11.9	764	4.3		412.3	950			553.0	770
Perodicticus potto	12.9	835	6.4	5.0	326.6	964				
Arctocebus calabarensis	7.2	309	4.4	3.6	131.2	206				
Nycticebus coucang	10.5	653	6.4	6.0	272.6	1160			609.2	1150
Nycticebus pygmaeus	7.2	307	6.1		171.9	470				
Loris tardigradus	6.1	193	5.5	5.6	107.4	167			196.8	159

The body mass values are species averages from compilations (sources noted below). The body mass values corresponding to RMR, intestinal mass, and intestinal surface area were taken from the specific individual(s) measured.

[a] Source: compilation by Isler et al. (2008) except data for *P. coquereli*, which were collected from specimens curated at the DLC.

[b] Source: compilations by Isler et al. (2008) and Gordon (2006) except data for *D. madagascariensis* and *P. coquereli*, which were collected by this study from animals at the DLC because data on wild-caught animals are not available.

[c] Source: compilation by Kappeler and Pereira (2003) and verified by DLC staff for those species housed by the facility.

[d] Source: compilation by Isler and van Schaik (2006b) and data collected by Hanna (2006), except for data from *D. madagascariensis*, *L. catta*, and *P. coquereli* which were collected by this study.

[e] Source: Campbell et al. (2000), Campbell (2003), and Campbell unpublished data. Values for *D. madagascariensis*, *L. catta*, and *P. coquereli* were supplemented by data collected by this study.

[f] Source: Chivers and Hladik (1980), Campbell et al. (2000) Campbell (2003), and Campbell unpublished data. Values for *D. madagascariensis*, *L. catta*, and *P. coquereli* were supplemented by data collected by this study.

mals is not available for the two of the species (*D. madagascariensis* and *P. coquereli*) that are the focus of this study. Body mass values were obtained from the DLC records for these species.

Gestation and weaning

Data on length of gestation and weaning age are provided in Table 1. These data were compiled in Kappeler and Pereira (2003), and most of which derive from records at the DLC. Age at weaning is difficult to determine because weaning is a gradual process rather than a precise landmark (Lee et al., 1991; Lee, 1996). The infants at the DLC are provided with food before they cease nursing, so some of the energetic resources are not derived from maternal sources. The adult foods have begun prior to this age, but weaning ages used in this study signify the cessation of maternal inputs and thus reflect maternal contributions to growth.

A few species included in the study often twin (*M. coquereli*, *M. murinus*, *V. variegata variegata*, *C. medius, G. demidovii*, *O. crassicaudatus*, and *Nycticebus pygmaeus*) and thus the maternal contribution is divided between two simultaneously gestating and subsequently nursing offspring. The length of gestation and the weaning age were divided by 2 in these species to account for the portion of the maternal contribution that would be devoted to a single offspring.

Resting metabolic rate

RMR and corresponding body mass measurements are provided in Table 1. Many of the data points were taken from the literature, as compiled by Isler and van Schaik (2006b) and Isler et al. (2008). RMR (also referred to as basal metabolic rate) reflects the basal level of energy required for maintenance of life, and does not include energy required for activities such as locomotion, feeding, growth, or reproduction. Specific methodological criteria, as outlined by Genoud (2002), must be met in order to ensure that only metabolic expenditure for the basic maintenance of existing tissues is measured. All measurements of RMR included in this study were taken from adult animals in a nonreproductive state (females were not pregnant or lactating) who were postabsorptive (not digesting food) at the time of data collection. They

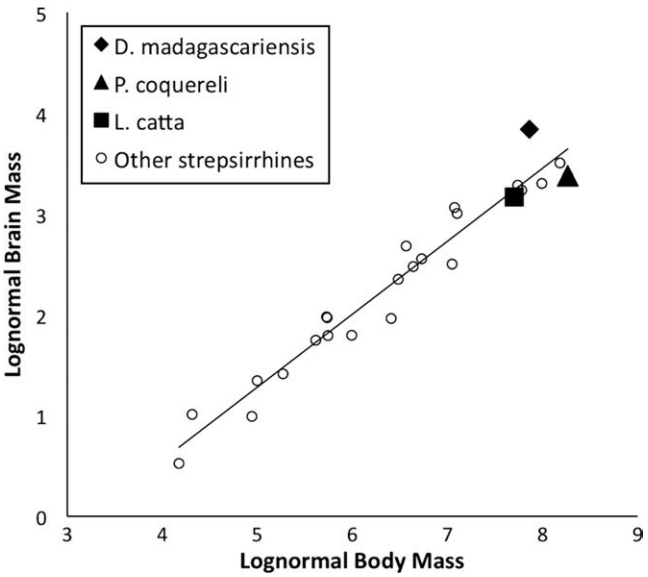


Fig. 1. Reduced-major-axis regression of log brain mass on log body mass for strepsirrhines ($y = -2.38 + 0.73x$, d.f. = 26, r-square = 0.95, $P < 0.0001$). The species in which growth curves were investigated for this study are highlighted.

rested in a chamber at an approximate thermoneutral temperature of 25–28°C. To ensure that the animals were not digesting food and in a lethargic state similar to their normal resting period, the animals were measured early in the morning before they had been fed. In the case of *Daubentonia*, the lighting is dimmed in their enclosures at about 11:00 am, stimulating their nocturnal activity, so these animals were measured during late morning.

Data on RMR was collected by this study for three species at the DLC (*D. madagascariensis*—3 males and 2 females; *P. coquereli*—3 males and 2 females; *L. catta*—3 males and 2 females), and the averages are presented in Table 1 along with corresponding averages of body mass. Oxygen consumption at rest was determined using the Sable System TR-1 package® (Sable Systems International; Las Vegas, NV), and details regarding the configuration of the chamber and equipment can be found in Hanna (2006). To calibrate the system and account for daily fluctuations in flow rate, nitrogen dilution was performed on the day of each data collection session following the methods developed by Fedak et al. (1981).

Intestinal size

Intestinal size refers to the mass and surface area of the small intestine, cecum, and colon. Values for intestinal mass and intestinal surface area were taken from published studies (Chivers and Hladik, 1980, with corrections in typesetting provided by K. Isler; Campbell et al., 2000; Campbell, 2003) and supplemented by data collected by this study. The gastrointestinal tracts that were measured for this study came from individuals who had died within the past 24 h, or had been frozen within 24 h of death. Measurements of mass and dimensions of one specimen of *D. madagascariensis* were taken before and after freezing; the values were comparable.

The methods utilized by each study were similar. Intestinal masses were taken after the contents were removed and do not include the mesentery. Intestinal surface area was determined by the following steps: the length and diameter of the small intestine, cecum, and colon were measured; the surface area of each structure was calculated by using the formula for the surface area of the sides of a cylinder ($2\pi r \times h$, where r = 1/2 of the diameter and h = length of structure); and the resulting surface areas of each structure were summed. Campbell (2003) and Campbell et al. (2000) did not always provide data on the diameter of the intestinal structures, so these values were determined by digitizing the scaled drawing provided in the publications.

TABLE 2. Subjects measured to assess cranial and somatic growth

Subject	Sex	Age at time of measurements
Daubentonia madagascariensis		
Sabrina	F	3 to 19 months
Medusa	F	3 to 19 months
Medea	F	1 to 8 months
Hitchcock	M	Neonate to 2 months
Morticia	F	Adult
Merlin	M	Adult
Kali	F	Adult
(cadaver)	U	Neonate
Lemur catta		
Nicea (twin)	F	2 to 14 months
Tellus (twin)	F	2 to 14 months
Nebuchadrezzer	F	2 to 14 months
Berisades	M	2 to 14 months
Ivy	M	Neonate to 12 months
Aracus	M	Adult
Cleis	F	Adult
Dory	F	Adult
Sosiphanes	F	Adult
(cadaver)	U	Neonate
(cadaver)	U	Neonate
(cadaver)	U	Neonate
(cadaver)	F	2 months
(cadaver)	M	3 months
(cadaver)	F	Adult
Propithecus coquereli		
Maximus	M	5 to 17 months
Geta	F	3 to 12 months
Marcus	M	2 to 15 months
Lucius	M	Neonate to 2 months
Maria	F	Neonate to 2 months
Alex	F	Adult
Nero	M	Adult
Martin	M	Adult
(cadaver)	F	Neonate
(cadaver)	F	1.5 months
(cadaver)	M	2 months
(cadaver)	M	Adult

M, male; F, female; U, unknown.

Analyses

Brain and body growth curves were generated by loess regression, which fits models to localized subsets of the distribution, building up a model that describes the entire bivariate distribution (Cleveland, 1979; Cleveland and Devlin, 1988). This curve-fitting technique has advantages over other methods such as Gompertz as it does not make any assumptions about the shape of the curve. It produces a curve with a minimized residual sum of squares, reflected in a high r-square value, for a given window or span of localized data. S-Plus was used to calculate the loess fits, which were "double-smoothed" to reduce any minor oscillations in the data points by

performing a second loess regression on the predicted values. Pseudo-velocity curves were derived from the growth curves to allow a detailed examination of changes in growth rate, similar to the method described in Coelho (1985) in which the difference in successive predicted mass values is divided by the difference in successive ages (see also Leigh, 2004; Leigh and Shea, 1995, 1996).

The analyses of data from adult animals, which analyzed brain maintenance and maternal energetic contributions, were performed on log-transformed means from each species. Evidence for trade-offs between life-history variables (gestation length and weaning age), metabolic rate, intestinal size, and encephalization were tested using two techniques. First, residuals were calculated from a reduced-major-axis regression of gestation length, weaning age, RMR, intestinal surface area, intestinal mass, and brain mass on their respective body mass values. Note that the variables of intestinal size and RMR for a given species had different corresponding body mass values because specific measurements were taken from an individual, or averaged from two or three individuals, rather than averaged from several animals. These residuals were then entered into a least-squares regression to determine whether any of the variables relative to body mass significantly predicts the degree of encephalization.

Second, the variables were analyzed using multiple regressions. The trade-offs may not be apparent when only one variable is examined relative to encephalization. Species may use various combinations of incremental decreases intestinal size and increases in RMR and life history to afford the cost of a substantial increase in encephalization. The regressions were performed with brain size as the dependent variable and body mass, life-history variables, intestinal surface area, and RMR as the independent variables. The body size variable used for these regressions was derived from a large sample of body mass measurements from males and female individuals of each species (Table 1). Only the measurement of intestinal surface area was used in the models because there were too few species with available data on both RMR and intestinal mass.

Phylogenetic nonindependence is a potentially confounding factor in any analysis of a large comparative dataset. The correlations between variables could be the result of inheritance from a common ancestor rather than evidence of a strictly adaptive relationship (Felsenstein, 1985). The calculation of independent contrasts is the most commonly used method to account for phylogenetic bias in continuous variables (Harvey and Pagel, 1991; Purvis and Webster, 1999; Nunn and Barton, 2001;). In this study, the PDAP module of the Mesquite program was used (Garland et al., 1999; Garland and Ives, 2000; Midford et al., 2003; Maddison and Maddison, 2007) to calculate independent contrasts for the variables that were analyzed in the regressions described above. The phylogeny used in this study derives from Purvis (1995), with updates from more recent studies by Smith and Cheverud (2002), Gordon (2006), and Horvath et al. (2008).

Martin et al. (2005) have raised some concerns about regarding the efficacy of using methods that attempt to correct for phylogenetic inertia; for instance, they argue that these methods may obscure relevant grade shifts and most of the inertia is accounted for by the variation in body mass. All of these potential problems are pertinent to this study, and therefore, all of the residuals and multiple regressions were calculated using both species values and independent contrasts. Any disparities in the level of significance between these two sets of analyses are noted in the results. Otherwise, only the results of the analyses of species values are reported.

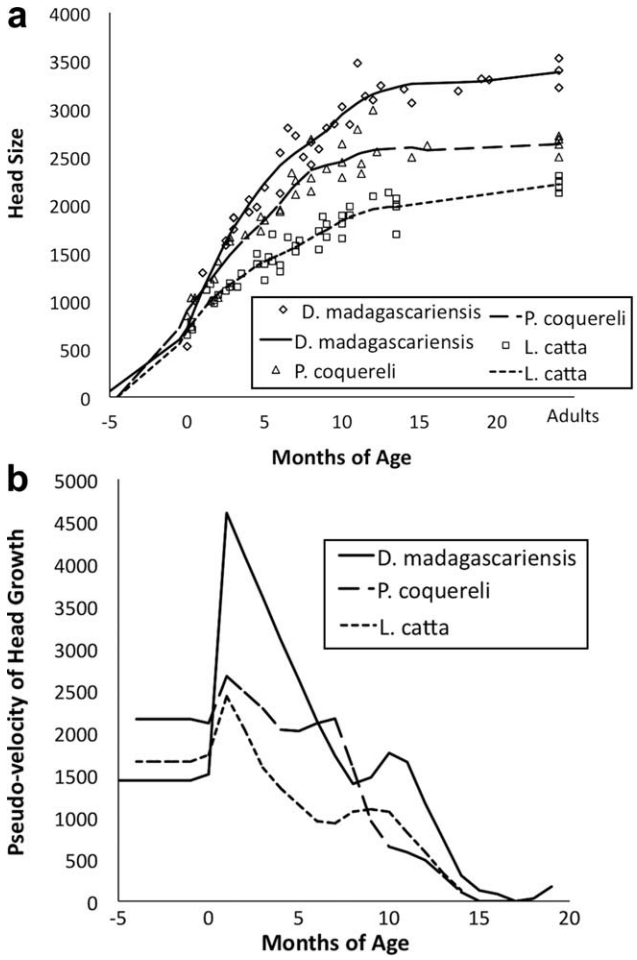


Fig. 2. Head growth trajectories in three lemur species: (**a**) individual measurements of head size (details regarding calculation are provided in the text) with smoothed loess regression; (**b**) pseudo-velocity curves for head growth, measured as head size gain per month and smoothed with a loess regression. Note that prenatal growth is inferred from length of gestation and neonatal brain size and was not measured directly. Details regarding the sample are provided in Table 2.

RESULTS

Growth rates

The relationship between brain and body size among the species included in this study is illustrated in Figure 1. Head growth was measured in three of these species, and the growth trajectories and pseudo-velocity curves for head size are provided in Figure 2. These results demonstrate that in these species, most of the brain growth occurs postnatally. In addition, the differences in brain size among the three species are attained by variation in growth rates rather than the duration of growth. The most encephalized and largest-brained species, *D. madagascariensis*, experiences the most rapid

brain growth of the three species during the first five months of ontogeny. The remaining species do not exhibit remarkably different brain growth trajectories, with *P. coquereli* only slightly outpacing *L. catta* during the first ten months of ontogeny. Thus, the hypothesis that *D. madagascariensis* is able to afford the energetic cost of growing a larger brain relative to other lemurs by extending the period of growth is not supported.

The body mass growth trajectories and pseudo-velocity curves are provided in Figure 3. We hypothesized that if *D. madagascariensis* experienced particularly rapid brain growth relative to *P. coquereli* and *L. catta*, then it would also experience a relatively low rate of body growth to compensate for the energetic costs of brain growth. The results demonstrate that this is not the case. The three species have similarly slow rates of prenatal body growth. Then *D. madagascariensis* experiences the most rapid body growth of the three species during the first two months of postnatal ontogeny, during which the most extensive brain growth is occurring. *P. coquereli* experiences an extended and pronounced growth spurt that continues until twelve months of age, while *L. catta* begins as a slow-growing neonate that increases its growth rate, reaching a peak at approximately five months of age. *L. catta* and *P. coquereli* do exhibit a more rapid body growth rate than *D. madagascariensis* from five months to seven months and from two months to twelve months, respectively. However, these time periods do not coincide with particularly rapid brain growth in *D. madagascariensis*. In order to determine how large-brained strepsirrhines are able to meet their energetic demands during growth, maternal contributions must be investigated.

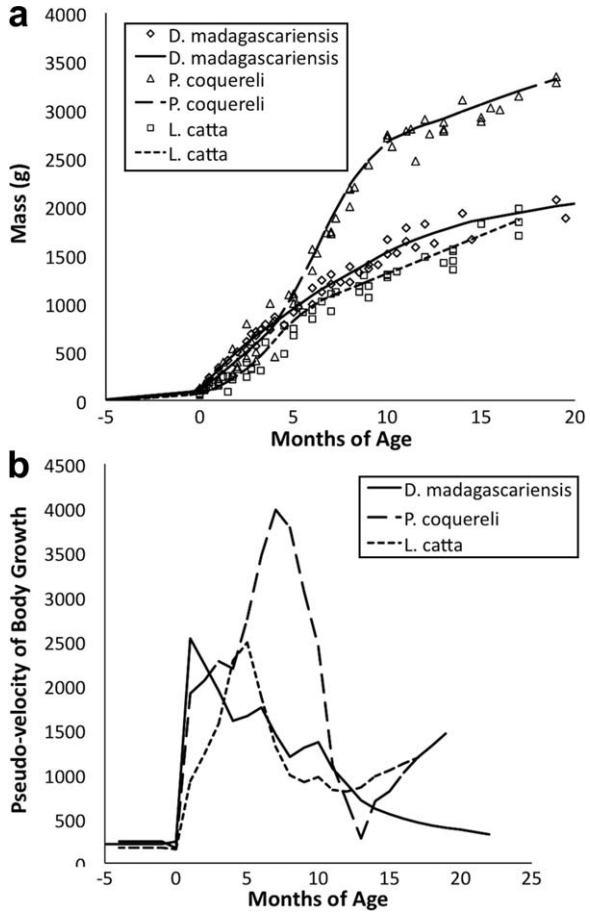


Fig. 3. Body mass growth trajectories in three lemur species: (**a**) individual measurements of body mass with smoothed loess regression; (**b**) pseudo-velocity curves for body growth, measured as body mass gain per month and smoothed with a loess regression. Details regarding the sample are provided in Table 2.

Life history: gestation length and weaning age

The results of the regressions between body mass and both gestation length and weaning age are provided in Table 3. The relationship between gestation length relative to body mass and encephalization is illustrated in Figure 4a, demonstrating that the length of the prenatal period has no effect on brain size. However, when lorisiforms are analyzed separately, the relationship is significant ($P = 0.05$). Weaning age relative to body mass is highly correlated with encephalization, as shown in Figure 4b. However, the analysis of independent contrasts does not reach significance ($P = 0.06$). Thus, there is only tentative support for the hypothesis that extension of life history contributes to the energetic demands of encephalization.

Resting metabolic rate

The relationship between body mass and RMR in the strepsirrhines is given in Table 3. Of the three species included in the analysis of growth rates, the adult *D. madagascariensis* has the highest RMR for its body mass, indicating that the mother is able to contribute a great deal of energy to her offspring. This result potentially explains how *D. madagascariensis* is able to afford the energetic cost of particularly rapid brain growth while maintaining a comparable body growth rate to that of *P. coquereli* and *L. catta*. However, this conclusion can be confirmed only with measurements of RMRs in gestating and lactating females.

A broader analysis of the relationship between encephalization and RMR relative to body mass, including several strepsirrhine species, is shown in Figure 5. A comparison of residuals generated from reduced-major-axis regressions of brain mass and RMR on body mass reveals that the RMR relative to the body mass predicts the degree of encephalization (Fig. 5a). However, the relationship is not strong, as demonstrated by an r-square of only 0.22. Similar results were found when

TABLE 3. Reduced-major-axis regressions of variables in study against body mass

	Intercept	Slope	Degrees of freedom	r-square	*P* value
Gestation length	−2.41	0.56	19	0.33	0.008
Weaning age	−2.26	0.52	15	0.32	0.02
RMR	0.92	0.73	18	0.89	<0.0001
Intestinal mass	−1.57	0.78	8	0.90	<0.0001
Intestinal surface area	1.18	0.76	17	0.89	<0.0001

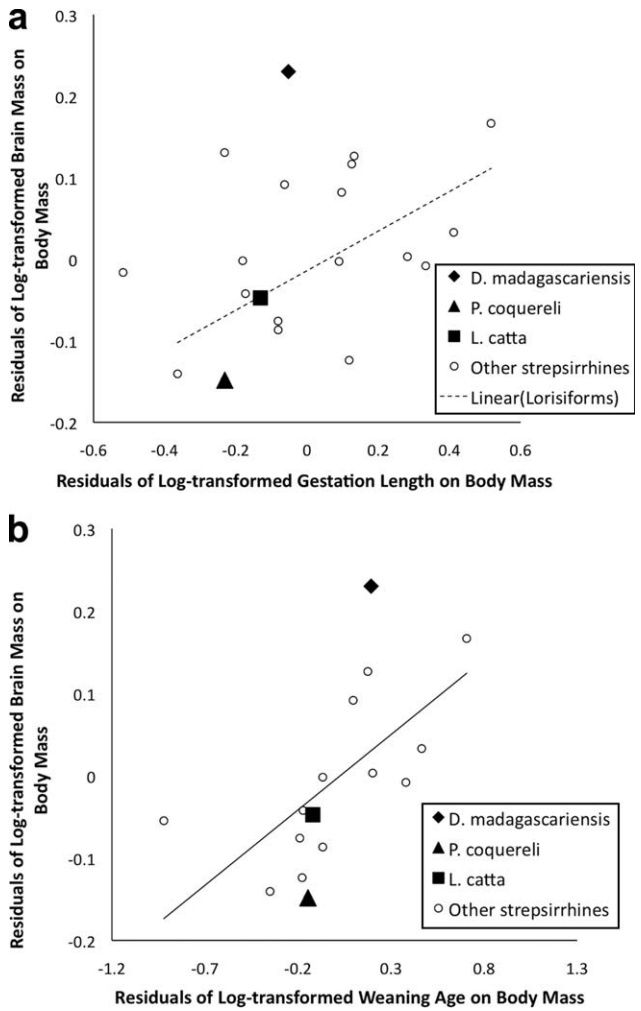


Fig. 4. Least-squares regressions of examining the relationship between life-history variables and encephalization. (**a**) Gestation length relative to body mass and encephalization in strepsirrhines is not significant ($y = 0.01 + 0.15x$, d.f. = 19, r-square = 0.13, $P = 0.11$), but is significant among lorisiforms ($y = -0.01 + 0.24$, d.f. = 7, r-square = 0.49, $P = 0.05$); (**b**) Relationship between weaning age relative to body mass and encephalization (species values: $y = -0.005 + 0.18x$, d.f. = 15, r-square = 0.38, $P = 0.01$; contrasts: intercept zeroed, 0.29x, d.f. = 14, r-square = 0.23, $P = 0.06$).

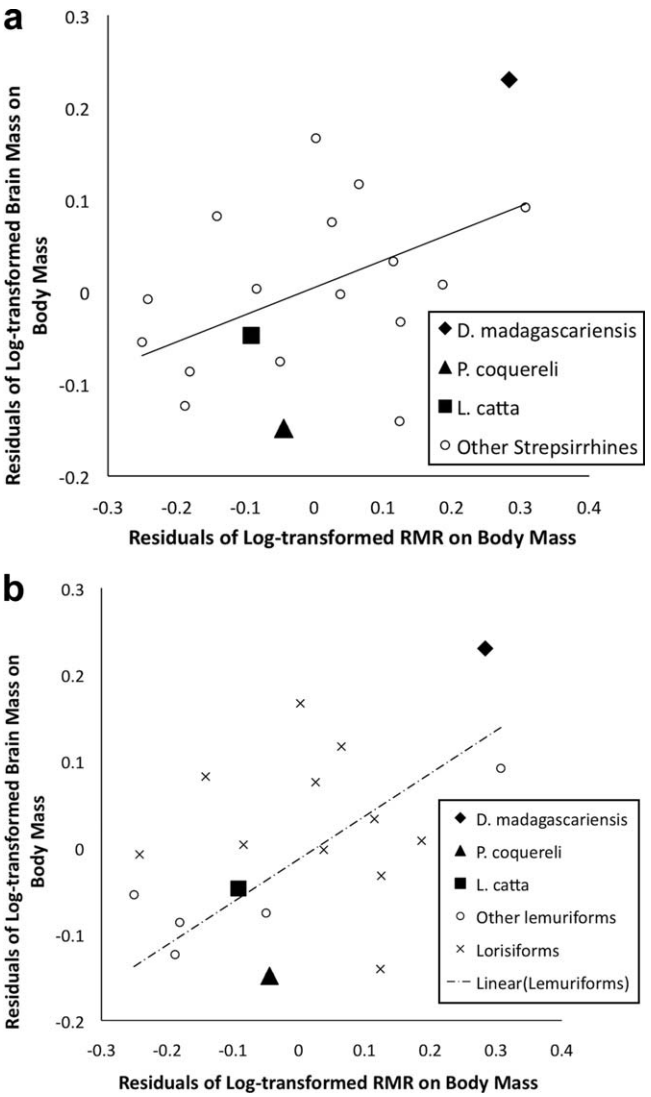


Fig. 5. Least-squares regression examining the relationship between RMR relative to body mass and encephalization. All regression lines are from analyses of species values. (**a**) The analysis including all of the strepsirrhines included in this study (species values: $y = -0.004 + 0.29x$, d.f. = 18, r-square = 0.22, $P = 0.04$; contrasts: intercept zeroed, 0.35x, d.f. = 17, r-square = 0.27, $P = 0.015$). B) Analyses of lemuriforms (species values: $y = -0.13 + 0.50x$, d.f. = 7, r-square = 0.69, $P = 0.01$; excluding *D. madagascariensis*: $y = -0.04 + 0.31x$, d.f. 6, r-square = 0.56, $P = 0.05$; contrasts: intercept zeroed, $x = 0.55$, d.f. = 7, r-square 0.72, $P = 0.004$) and lorisiforms (species values: $y = 0.03 - 0.13x$, d.f. = 10, r-square = 0.04, $P = 0.54$; contrasts: intercept zeroed, $x = -0.19$, d.f. = 9, r-square 0.12, $P = 0.3$).

independent contrasts were analyzed instead of species values. However, when the lemuriforms and lorisiforms are analyzed separately, the results reveal a strong positive relationship between RMR relative to body mass and encephalization in lemuriforms but not in lorisiforms (Fig. 5b). Analyses of independent contrasts produce similar results. In addition, the analysis remains significant with the exclusion of *D. madagascariensis*, whose large brain size and high RMR may drive the correlation.

Intestinal size

Two measures of the intestines are used to analyze the relationship between intestinal size and encephalization: mass and surface area. The results of reduced-major-axis regressions of intestinal mass, intestinal surface area, and corresponding body mass data is presented in Table 3.

An extensive analysis of the relationship between encephalization and intestinal size in a larger sample of strepsirrhines produces mixed results depending on which measure of intestinal size is examined. The relationship between intestinal mass relative to body mass and encephalization is negative and significant (Fig. 6a). The sample size was insufficient to analyze lemuriforms and lorisiforms separately, and *D. madagascariensis* does not appear to be an outlier that is driving this relationship.

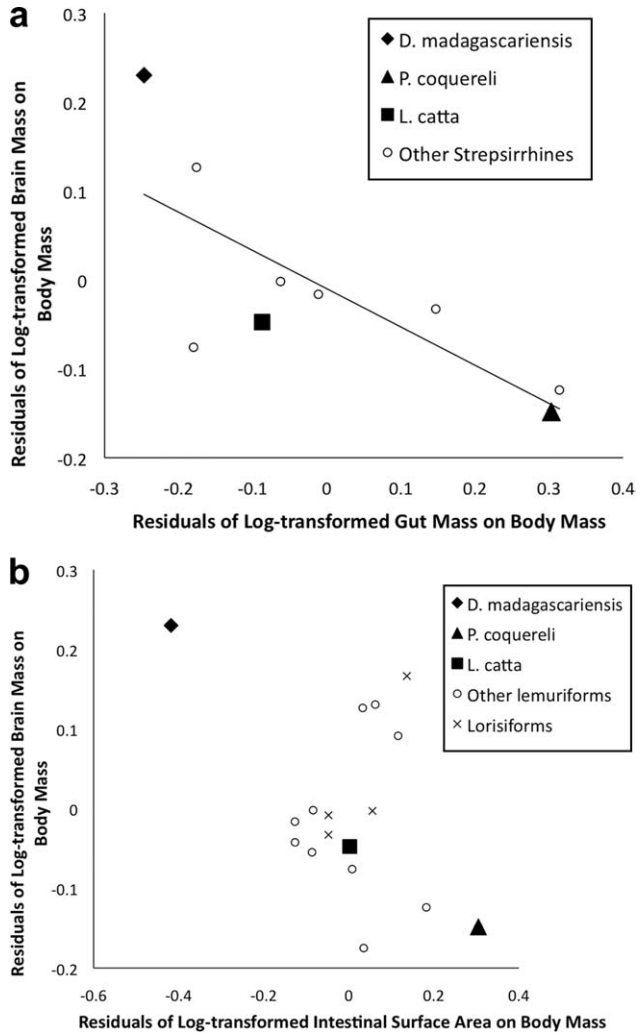


Fig. 6. Least-squares regressions examining the relationship between intestinal size relative to body mass and encephalization. (**a**) Analysis of intestinal mass and encephalization ($y = -0.01 - 0.43x$, d.f. = 8, r-square = 0.56, $P = 0.02$; contrasts: intercept zeroed, $-0.27x$, d.f. = 7, r-square = 0.42, $P = 0.04$). (**b**) Analysis of intestinal surface area and encephalization (strepsirrhines: $y = 0.0009 - 0.28x$, d.f. = 18, r-square = 0.15, $P = 0.12$; lemuriforms: $y = -0.01 - 0.37x$, d.f. = 12, r-square = 0.29, $P = 0.057$; contrasts of strepsirrhines: intercept zeroed, -0.31, d.f. = 17, r-square = 0.19, $P = 0.08$).

The relationship between intestinal surface area relative to body mass and encephalization is negative and but not significant (Fig. 6b). If lemuriforms are analyzed separately, the relationship is close to significant ($P = 0.057$). It appears that *D. madagascariensis* is driving this trend; when that species is removed, the relationship is not significant ($P = 0.51$). However, a cluster of three lemuriforms with high degrees of encephalization and large intestinal surfaces areas (*E. mongoz*, *E. coronatus*, and *H. griseus*) are also shifting the regression line. *E. mongoz* has a high RMR, which may compensate for the energetic needs of its large intestinal size and brain size; no data are available on RMR in *E. coronatus* and *H. griseus*. The analyses of independent contrasts of strepsirrhines and lemuriforms produce similar results. Lorisiforms were not analyzed separately because of insufficient sample size.

In sum, there is some support for the hypothesis that some taxa reduce their intestinal size in order to provide additional energy for brain maintenance and the maternal contribution to brain growth.

Multiple regression: encephalization, life history, energetics, and expensive tissues

The multiple regressions examined the relationship between brain mass and four effects: body mass, weaning age, RMR, and intestinal surface area. Intestinal mass was not included in the analyses because of insufficient sample size. The results of the analyses are provided in Table 4. Among both strepsirrhines and lemuriforms, weaning age did not have a significant effect on brain mass when the other variables were included in the analysis. Intestinal surface area had a significant negative effect among both strepsirrhines and lemuriforms, and RMR had a significant positive effect only in lemuriforms.

Intestinal surface area and RMR had the strongest effects, particularly among lemuriforms, so another analysis was performed including only these independent variables. The results are reported in Table 4. Among strepsirrhines, neither variable significantly affected brain mass, though analyses of independent contrasts did produce significant results. These results could be due, in part, to the different relationships among lemuriforms and lorisiforms, respectively. In contrast, both variables had a significant effect among lemuriforms, but the analysis of independent contrasts produced nonsignificant results. It is not clear why the disparity is present, though it is important to note that the sample size of contrasts was small ($n = 6$). These analyses included only a sample of species with data available on both intestinal size and RMR, and the currently available sample size is small. Therefore, testing models with several independent effects is difficult and the results are tentative.

DISCUSSION AND CONCLUSIONS

This study tested the predictions of the expensive tissue hypothesis (Aiello and Wheeler, 1995), the maternal energy hypothesis (Martin 1996, 1998; Martin and Isler, in press), and the expensive brain hypothesis (Isler and van, Schaik 2006a, 2009) in strepsirrhines, and found substantial support for the notion that brain size is constrained by energetic costs. Some trade-offs must be made to evolve increased encephalization. The analyses of the ontogenetic period did not reveal a trade-off between brain and body growth in the offspring, but the analyses of weaning age, as well as RMR and intestinal size in adults, did reveal trade-off in maternal support of the growing offspring.

Daubentonia madagascariensis is the most encephalized of the strepsirrhines, and we expected that it would either increase the duration of its brain growth or decrease its body growth rate to compensate for the energy required to grow a large brain. Neither of these results was found. *Daubentonia* experiences a remarkably rapid brain growth and no simultaneous decrease in body growth rate as compared to *L. catta*. Instead, the energy needed to grow a large brain may come from maternal sources.

Postnatal maternal investment is a critical factor when considering the reproductive strategies, and this

TABLE 4. Results of multiple regressions examining the tradeoffs between body mass, life history, intestinal surface area, and RMR on brain mass

Sample	Dependent variable	Independent variable	Coefficient	Standard error	*P*-value
Strepsirrhines	Brain mass	Body mass	0.82	0.13	0.004[a]
$n = 9$		Weaning age	0.17	0.07	0.08
		RMR	0.14	0.14	0.37
		Intestinal surface area	−0.38	0.11	0.03[a]
Lemuriforms	Brain mass	Body mass	0.68	0.05	0.006[a]
$n = 7$		Weaning age	−0.10	0.06	0.24
		RMR	0.48	0.08	0.03[a]
		Intestinal surface area	−0.35	0.04	0.01[a]
Strepsirrhines	Brain mass	Body mass	0.76	0.21	0.002[a]
$n = 11$		RMR	0.21	0.19	0.31[a]
		Intestinal surface area	−0.25	0.16	0.16[a]
Lemuriforms	Brain mass	Body mass	0.73	0.06	0.0007[a]
$n = 7$		RMR	0.36	0.06	0.007[b]
		Intestinal surface area	−0.35	0.05	0.005[b]

[a] Significant results were found when independent contrasts were analyzed.
[b] Significant results were not found when independent contrasts were analyzed.

study suggests that encephalized lemuriforms are able to meet the energetic costs primarily by increasing their RMR. Weaning age is later in encephalized lemurs, but brain growth, at least in *Daubentonia*, is complete prior to weaning. Thus, the higher RMR in *Daubentonia* relative to other lemurs provides a more logical explanation. However, the measures of RMR analyzed in this study derive from nonreproducing adults, so it is not entirely clear whether energetics play a significant role in developmental rates and durations, and the degree of encephalization. In addition, this study demonstrates correlations between these variables, but cannot establish a causative effect without more closely examining reproduction and ontogeny. Studies of RMR in infants and lactating females would further clarify the variation in energetic investment among this taxon. In particular, we would expect *Daubentonia* mothers and offspring to have a particularly elevated RMR.

When a larger sample of strepsirrhines was investigated, the results revealed that there are energetic tradeoffs that may explain how mothers are able to afford encephalized offspring. However, it is important to note that the specific trade-offs depend on the taxon. Encephalized lemuriforms most likely rely on high maternal energetic turnover, as indicated by high relative RMR. In addition, this study found some support for the hypothesis that encephalized lemuriforms have a smaller intestinal size, allowing more energy to be funneled towards offspring. Also, encephalized offspring with small intestinal size are able to channel more energy towards brain growth rather than intestinal growth. For instance, an immature *L. catta* must grow a gut that is 50% larger the gut of *Daubentonia*. In contrast, encephalized lorisiforms have an extended period of gestation and lactation but no alterations to intestinal size or RMR.

Studies have found that other expensive tissues, such as pectoral mass in birds (Isler and van Schaik, 2006a), testicles size in bats (Pitnick et al., 2006), and gut size in fish (Kaufman, 2003), are reduced in species with large brain masses, suggesting that there is an energetic trade-off between these tissues. However, particularly in regards to gut size, there are other confounding factors. Dietary quality is another variable that has shown a strong positive relationship to brain mass (Fish and Lockwood, 2003), though this measure is probably a direct reflection of gut size. McNab (2002) notes that primates with low quality diets tend to have a large gut to process the highly fibrous material and a low metabolic rate to conserve energy. These factors may have little to do with brain size directly, but they would certainly constraint a species ability to direct energy towards brain growth and maintenance.

It is interesting to note that Maclean et al. (2009) found that a higher quality diet, specifically frugivory relative to folivory, is associated with greater degrees of encephalization among lemuriforms. In contrast, they found no relationship between encephalization and group size or pairbondedness among the lemuriforms, suggesting that some combination of foraging skill and energetic constraints might better explain the variation in relative brain size rather than the benefits to sociality. Furthermore, Kaufman et al. (2005) compared the brain morphology of the highly encephalized *Daubentonia* to that of other, less-encephalized strepsirrhines, and they found a particularly large frontal lobe in the former. They suggest that this morphology reflects the cognitively complex foraging behavior and sensorimotor demands of acquiring high-quality food in *Daubentonia*. In turn, these high quality foods, easily digested with their small intestinal size, could provide the necessary energy for their large brains.

This study demonstrated that the evolution of a large brain requires some sort of compensatory mechanism to meet the energetic demands. It is important to note that these mechanisms take many forms and different strategies have evolved across taxa. This analysis of strepsirrhines revealed a distinction between lemuriforms and lorisiforms. The results are most interesting in lemuriforms because the highly encephalized *Daubentonia* was shown to have a remarkably high metabolic rate and a very small intestinal size. In many ways, *Daubentonia* is the human of the lemurs, and the energetic trade-offs are most salient in these instances.

ACKNOWLEDGMENTS

The authors greatly appreciate the support of the all of the staff at the Duke Lemur Center, especially the efforts of Sarah Zehr and David Brewer. Christine Drea, Leslie Digby, and Matt Cartmill provided valuable mentorship during the project. This manuscript was considerably improved by the comments provided by Karin Isler, Robert D. Martin, and Leslie Aiello. This study was approved by the Institutional Animal Care and Use

Committee, protocols A393-03-12 and A347-06-10, and Duke Lemur Center Research #M12-03-1 and #M-10-06-3. This study is archived at the Duke Lemur Center with the publication number DLC #1174.

LITERATURE CITED

Aiello LC, Bates N, Joffe T. 2001. In defence of the expensive tissue hypothesis. In: Falk D, Gibson KR, editors. Evolutionary anatomy of the primate cerebral cortex. Cambridge: Cambridge University Press. p 57–78.

Aiello LC, Wheeler P. 1995. The expensive-tissue hypothesis. Curr Anthropol 36:199–221.

Allman J. 1995. Brain and life span in Catarrhine primates. In: Butler R, Brody J, editors. Delaying the onset of late-life dysfunction. New York: Springer. p 221–241.

Allman J, Hasenstaub A. 1999. Brains, maturation times, and parenting. Neurobiol Aging 20:447–454.

Allman J, McLaughlin T, Hakeem A. 1993. Brain weight and lifespan in primate species. Proc Natl Acad Sci USA 90:3359–3563.

Armstrong E. 1983. Relative brain size and metabolism in mammals. Science 220:1302–1304.

Armstrong E. 1985. Relative brain size in monkeys and prosimians. Am J Phys Anthropol 66:263–273.

Austad S, Fischer K. 1992. Primate longevity: its place in the mammalian scheme. Am J Primatol 28:251–261.

Barrickman NL. 2008. Evolutionary relationship between life history and brain growth in anthropoid primates. Durham, NC: Duke University.

Barrickman NL, Bastian ML, Isler K, van Schaik CP. 2008. Life history costs and benefits of increased brain size: a comparative test using primates. J Hum Evol 54:568–590.

Barton RA. 1996. Neocortex size and behavioural ecology in primates. Proc R Soc B Biol Sci 263:173–177.

Barton RA. 1999. The evolutionary ecology of the primate brain. In: Lee P, editor. Comparative primate socioecology. Cambridge: Cambridge University Press. p 167–194.

Bauchinger U, Wohlmann A, Biebach H. 2005. Flexible remodeling of organ size during spring migration of the garden warbler (*Sylvia borin*). Zoology 108:97–106.

Bauchot R. 1978. Encephalization in vertebrates—new mode of calculation for allometry coefficients and isoponderal indexes. Brain Behav Evol 15:1–18.

Byrne RW, Corp N. 2004. Neocortex size predicts deception rate in primates. P Roy Soc B Biol Sci 271:1693–1699.

Byrne RW, Whiten, A. 1988. Machiavellian intelligence: social expertise and the evolution of intellect in monkeys, apes, and humans. Oxford, UK: Clarendon.

Cabana T, Jolicoeur P, Michaud J. 1993. Prenatal and postnatal growth and allometry of stature, head circumference, and brain weight in Quebec children. Am J Hum Biol 5:93–99.

Campbell J. 2003. Malagasy primates: nutrition and gastrointestinal physiology with an emphasis on fiber utilization. Raleigh, NC: North Carolina State University.

Campbell J, Eisemann J, Williams C, Glenn K. 2000. Description of the gastrointestinal tract of five lemur species: *Propithecus tattersali, Propithecus verreauxi coquereli, Varecia variegata, Hapalemur griseus,* and *Lemur catta*. Am J Primatol 52:133–142.

Chivers DJ, Hladik CM. 1980. Morphology of the gastrointestinal tract in primates: comparisons with other mammals in relation to diet. J Morphol 166:337–386.

Cleveland W. 1979. Robust locally weighted regression and smoothing scatterplots. J Am Stat Assoc 74:829–836.

Cleveland W, Devlin S. 1988. Locally-weighted regression: an approach to regression analysis by local fitting. J Am Stat Assoc 83:596–610.

Coelho M. 1985. Baboon dimorphism: growth in weight, length, and adiposity. In: Watts E, editor. Nonhuman primate models for human growth and development. New York: Alan R. Liss. p 125–159.

Deaner R, Barton R, van Schaik CP. 2003. Primate brain and life histories: renewing the connection. In: Kappeler P, Pereira M, editors. Primate life histories and socioecology. Chicago: University of Chicago Press.

Deaner RO, Isler K, Burkart J, van Schaik CP. 2007. Overall brain size, and not encephalization quotient, best predicts cognitive ability across non-human primates. Brain Behav Evol 70:115–124.

Dunbar RIM. 2003. The social brain: mind, language, and society in evolutionary perspective. Annu Rev Anthropol 32:163–181.

Dunbar RIM, Shultz S. 2007. Understanding primate brain evolution Philos Trans R Soc B 362:649–658.

Eisenberg JF, Wilson DE. 1978. Relative brain size and feeding strategies in the Chiroptera. Evolution 32:740–751.

Epstein HT, Epstein EB. 1978. Relationship between brain weight and head circumference from birth to age 18 years. Am J Phys Anthropol 48:471–473.

Falk D. 1987. Hominid paleoneurology. Annu Rev Anthropol 16:13–30.

Fedak MA, Rome L, Seeherman HJ. 1981. One-Step N2-dilution technique for calibrating open-circuit Vo2 measuring systems. J Appl Physiol 51:772–776.

Felsenstein J. 1985. Phylogenies and the comparative method. Am Nat 125:1–15.

Fish JL, Lockwood CA. 2003. Dietary constraints on encephalization in primates. Am J Phys Anthropol 120:171–181.

Foley RA, Lee PC. 1991. Ecology and energetics of encephalization in hominid evolution. Philos Trans R Soc B 334:223–232.

Garland T, Ives A. 2000. Using the past to predict the present: confidence intervals for regression equations in phylogenetic comparative methods. Am Nat 155:346–364.

Garland T, Midford P, Ives A. 1999. An introduction to phylogenetically-based statistical methods, with a new method for confidence intervals on ancestral states. Am Zool 39:374–388.

Genoud M. 2002. Comparative studies of basal rate of metabolism in primates. Evol Anthropol 11:108–111.

Gordon AD. 2006. Scaling of size and dimorphism in primates II: macroevolution. Int J Primatol 27:63–105.

Hakeem A, Sandoval G, Jones M, and Allman J. 1996. Brain and life span in primates. In: Birren J, Schaie K, editors. The handbooks of aging; handbook of the psychology of aging, 4th ed. San Diego: Academic Press. p 78–104.

Hanna J. 2006. Climbing energetics in primates: implications for primate locomotor evolution. Durham, NC: Duke University.

Harvey PH, Clutton-Brock TH. 1985. Life history variation in primates. Evolution 39:559–581.

Harvey PH, Martin RD, Clutton-Brock TH. 1987. Life histories in comparative perspective. In: Smuts B, Cheney D, Seyfarth R, Wrangham R, Struhsaker T, editors. Primate societies. Chicago: University of Chicago Press. p 181–196.

Harvey PH, Pagel M. 1991. The comparative method in evolutionary biology. Oxford: Oxford University Press.

Hladik CM, Chivers DJ, Pasquet P. 1999. On diet and gut size in non-human primates and humans: is there a relationship to brain size? Discussion and criticism. Curr Anthropol 40:695–697.

Hofman M. 1983. Evolution of brain size in neonatal and adult placental mammals—a theoretical approach. J Theor Biol 105:317–332.

Hofman M. 1993. Encephalization and the evolution of longevity in mammals. J Evol Biol 6:209–277.

Holliday M. 1986. Body composition and energy needs during growth. In: Faulkner F, Tanner J, editors. Human growth: a comprehensive treatise, 2nd ed. New York: Plenum Press. p 101–117.

Horvath J, Weisrock D, Embry S, Fiorentino I, Balhoff J, Kappeler P, Wray G, Willard H, Yoder A. 2008. The development and application of a phylogenomic toolkit: resolving the evolutionary history of Madagascar's lemurs. Genome Res 18:489–499.

Isler K, Kirk EC, Miller JMA, Albrecht GA, Gelvin BR, Martin RD. 2008. Endocranial volumes of primate species: scaling analyses using a comprehensive and reliable dataset. J Hum Evol 55:967–978.

Isler K, van Schaik CP. 2006a. Costs of encephalization: the energy trade-off hypothesis tested on birds. J Hum Evol 51:228–243.

Isler K, van Schaik CP. 2006b. Metabolic costs of brain size evolution. Biol Letters 2:557–560.
Isler K, van Schaik CP. 2009. The Expensive Brain: a framework for explaining evolutionary changes in brain size. J Hum Evol 57:392–400.
Judge D, Carey J. 2000. Postreproductive life period predicted by primate patterns. J Gerontol Biol Sci A 55:B201–B209.
Kappeler PM. 1996. Causes and consequences of life-history variation among strepsirhine primates. Am Nat 148:868–891.
Kappeler PM, Pereira ME. 2003. Primate life histories and socioecology. Chicago: University of Chicago Press.
Karasov W, Pinshow B, Starck J, Afik D. 2004. Anatomical and histological changes in the alimentary tract of migrating blackcaps (*Sylvia atricapilla*): a comparison among fed, fasted, food-restricted, and refed birds. Physiol Biochem Zool 77:149–160.
Kaufman J. 2003. On the expensive-tissue hypothesis: independent support from highly encephalized fish. Curr Anthropol 44:705–707.
Kaufman J, Ahrens E, Laidlaw D, Zhang S, Allman J. 2005. Anatomical analysis of an aye-aye brain (*Daubentonia madagascariensis*. Primates: Prosimii) combining histology, structural magnetic resonance imaging, and diffusion-tensor imaging. Anat Rec A 287:1026–1037.
Kety S, editor. 1957. The general metabolism of the brain in vivo. New York: Pergammon. p 221–237.
Kety S, Schmidt C. 1948. The nitrous oxide method for the quantitative determination of cerebral blood flow in man: theory, procedure and normal values. J Clin Invest 27:476–483.
Lee PC. 1996. The meanings of weaning: growth, lactation, and life history. Evol Anthropol 5:87–98.
Lee PC, Majluf P, Gordon I. 1991. Growth, weaning and maternal investment from a comparative perspective. J Zool 225:99–114.
Leigh SR. 2004. Brain growth, life history and cognition in primate and human evolution. Am J Primatol 62:139–164.
Leigh SR, Shea BT. 1995. Ontogeny and the evolution of adult body size dimorphism in apes. Am J Primatol 36:37–60.
Leigh SR, Shea BT. 1996. Ontogeny of body size variation in African apes. Am J Phys Anthropol 99:43–65.
Leonard WR, Robertson ML. 1992. Nutritional requirements and human evolution: a bioenergetics model. Am J Hum Biol 4:179–195.
Leonard WR, Robertson ML. 1994. Evolutionary perspectives on human nutrition. The influence of brain and body size and diet and metabolism. Am J Hum Biol 6:77–88.
Leonard WR, Robertson ML, Snodgrass J, Kuzawa C. 2003. Metabolic correlates of hominid brain evolution. Comp Biochem Physiol A 136:5–15.
Mace GM, Harvey PH, Clutton-Brock TH. 1981. Brain size and ecology in small mammals. J Zool 193:333–354.
Maclean EL, Barrickman NL, Johnson EM, Wall CE. 2009. Sociality, ecology, and relative brain size in lemurs. J Hum Evol 56:471–478.
Maddison W, Maddison D. 2007. Mesquite: a modular system for evolutionary analysis. Version 2.01. Available at:http://mesquiteproject.org.
Martin RD. 1981. Relative brain size and basal metabolic rate in terrestrial vertebrates. Nature 293:57–60.
Martin RD. 1983. Human brain evolution in an ecological context. 52nd James Arthur Lecture on the Evolution of the Human Brain. New York: American Museum of Natural History.
Martin RD. 1990. Primate origins and evolution: a phylogenetic reconstruction. Princeton, NJ: Princeton University Press.
Martin RD. 1996. Scaling of the mammalian brain: the maternal energy hypothesis. News Physiol Sci 11:149–156.
Martin RD. 1998. Comparative aspects of human brain evolution: scaling, energy costs, and confounding variables. In: Jablonski N, Aiello L, editors. The origin and diversification of language. San Francisco: California Academy of Sciences.
Martin RD. 2007. The evolution of human reproduction: a primatological perspective. Yearb Phys Anthropol 50:59–84.
Martin RD, Genoud M, Hemelrijk C. 2005. Problems of allometric scaling analysis: examples from mammalian reproductive biology. J Exp Biol 208:1731–1747.
Martin RD, Isler K. in press. The maternal energy hypothesis of brain evolution: an update. In: Broadfield D, Yuan M, Toth N, Schick K, editors. The human brain evolving: papers in honor of Ralph L Holloway. Bloomington, IN: Stone Age Institute Press.
McNab BK. 2002. The physiological ecology of vertebrates. Ithaca, NY: Cornell University Press.
McNab BK, Eisenberg J. 1989. Brain size and its relation to the rate of metabolism in mammals. Am Nat 133:157–167.
Midford P, Garland T, Maddison W. 2003. PDAP Package.
Milton K. 1981. Distribution patterns of tropical plant foods as an evolutionary stimulus to primate mental development. Am Anthropol 83:534–548.
Mink JW, Blumenschine RJ, Adams DB. 1981. Ratio of central nervous system to body metabolism in vertebrates—its constancy and functional basis. Am J Physiol 241:R203–R212.
Nunn CL, Barton RA. 2001. Comparative methods for studying primate adaptation and allometry. Evol Anthropol 10:81–98.
Pagel M, Harvey PH. 1988. The taxon-level problem in the evolution of mammalian brain size—facts and artifacts. Am Nat 132:344–359.
Pitnick S, Jones KE, Wilkinson GS. 2006. Mating systems and brain size in bats. Proc R Soc B Biol Sci 273:719–724.
Purvis A. 1995. A composite estimate of primate phylogeny. Philos Trans R Soc B 348:405–421.
Purvis A, Webster A. 1999. Phylogenetically independent comparisons and primate phylogeny. In: Lee P, editor. Comparative primate socioecology. Cambridge: Cambridge University Press.
Rasmussen D, Izard M. 1988. Scaling of growth and life history traits relative to body size, brain size, and metabolic rate in Lorises and Galagos (Lorisdiae. Primates). Am J Phys Anthropol 75:357–367.
Reader SM, Laland KN. 2002. Social intelligence, innovation, and enhanced brain size in primates. Proc Natl Acad Sci USA 99:4436–4441.
Ross C. 2003. Life history, infant care strategies, and brain size in primates. In: Kappeler P, Pereira M, editors. Primate life histories and socioecology. Chicago: University of Chicago Press. p 266–284.
Ross C. 2004. Life histories and the evolution of large brain size in great apes. In: Russon A, Begun D, editors. The evolution of thought: evolutionary origins of Great Ape intelligence. Cambridge: Cambridge University Press. p 122–139.
Ross C, Jones K. 1999. Socioecology and the evolution of primate reproductive rates. In: Lee P, editor. Comparative primate socioecology. Cambridge: Cambridge University Press. p 73–110.
Sacher G. 1978. Longevity, aging, and death: an evolutionary perspective. Gerontol 18:112–119.
Sacher G, Staffeldt E. 1974. Relation of time to brain weight for placental mammals: implications for the theory of vertebrate growth. Am Nat 108:593–615.
Sawaguchi T. 1988. Correlations of cerebral indices for 'extra' cortical parts and ecological variables in primates. Brain Behav Evol 32:129–140.
Sawaguchi TK, Kudo H. 1990. Neocortical development and social-structure in primates. Primates 31:283–289.
Shultz S, Dunbar RIM. 2007. The evolution of the social brain: anthropoid primates contrast with other vertebrates. Proc R Soc B Biol Sci 274:2429–2436.
Skoyles J. 2008. Human metabolic adaptations and prolonged expensive neurodevelopment: a review. Available from Nature Proceedings <http://hdlhandlenet/10101/npre200818562>.
Smith RJ, Cheverud JM. 2002. Scaling of sexual dimorphism in body mass: a phylogenetic analysis of Rensch's rule in primates. Int J Primatol 23:1095–1135.
Sol D, Bacher S, Reader SM, Lefebvre L. 2008. Brain size predicts the success of mammal species introduced into novel environments. Am Nat Suppl 1:S63–S71.
Stephan H, Frahm H, Baron G. 1981. New and revised data on volumes of brain structures in insectivores and primates. Folia Primatol 35:1–29.
Vinicius L. 2005. Human encephalization and developmental timing. J Hum Evol 49:762–776.